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Documentation (AWS Project).

1. Network design:-

- This network is designed on top of AWS VPC (virtual private cloud), where resources are divided into separate subnets based on their security and availability requirements. The overall goal of the design is simple.
- Two main subnets have been created inside the VPC: a public subnet and a private subnet. The public subnet ~~and a private~~ hosts components such as the application/web server and load balancer which are directly accessible from internet.
 - The private subnet host the database server, which can be accessed by the application server - there is no direct connection from the internet to this subnet.

2. Why the Database is kept in Private Subnet?

→ Security:- The risk of direct external attacks (hacking attempts, brute force, malware) on the database drops significantly, since no public IP is ever assigned to it.

→ Controlled access:- Only the application server (located in public subnet) can connect to the database, through a specific port and security group rule.



→ This architecture therefore gives the database an extra layer of protection - an attacker cannot reach the database without first bypassing the application server, which significantly strengthens the overall security posture.

3. Purpose of the NAT Gateway :-

→ The NAT (Network Address Translation) Gateway is used to ~~be~~ give private subnet resources outbound access to the internet, without making them accessible from the internet inbound.

• One way traffic :- The NAT Gateway only allows outbound (private → internet) requests; inbound (internet → private) request remains ~~un~~ blocked.

• High availability :- AWS NAT Gateway is a fully managed service that scales automatically and remains highly available, without requiring extra maintenance.

4. Purpose of Internet Gateway (IGW)

→ The IGW is VPC component that connects the VPC to the public internet. It allows bidirectional communication for resources placed in the public subnet (such as the web server or load balancer) - meaning users



on the internet can send request, and the server can also send responses back to them.

→ In the route table, the default route (0.0.0.0/0) for the public subnet point to the IGW, allowing public subnet traffic to reach the internet directly.

5. How the VPN provides secure connectivity:-

→ The VPN (Virtual Private Network) in this architecture is used to give administrators and company employees secure, encrypted access to private VPC resources, without ever exposing those resources to the public internet. Through a site-to-site VPN or client VPN, an encrypted tunnel is established between the company's on premises network/user device and the AWS VPC.

• Private access:- Users connecting through the VPN can securely access private subnet resources (such as the database on an internal admin panel), which normal internet user cannot reach at all.

• Authentication:- Proper authentication (certificate/credential) is verified before a VPN connection is established, reducing the risk of unauthorised access.



6. Challenges faced and How they were Resolved:-

→ Challenge 1: Private subnet resources had no internet access:-

⇒ A NAT Gateway entry was added to the private subnet's route table, allowing outbound traffic to reach the internet through the NAT Gateway, while inbound access remained blocked.

→ Connectivity between the application server and database server was failing:-

⇒ Security Group rules were corrected - the database security group was configured to allow inbound traffic only from the application server's security group, on specific port:

→ Some private resources were still not accessible even after connecting the VPN:-

⇒ Missing static routes were added between the VPN's customer gateway and the VPC route tables, so that VPN traffic could be routed to the correct subnet.